Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

If an air conditioner is put in the middle of a room and started working

(A) the room can be cooled slightly

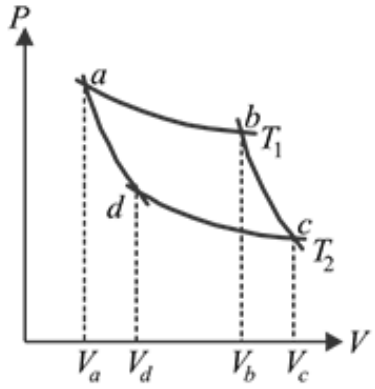
(B) the temperature of the room will not change

(C) the room will become slightly warmer

(D) the same temperature will be attained in the room as by putting it on the window in the standard position

Q.2

In the following $P - V$ diagram two adiabatics cut two isothermals at temperatures T_1 and T_2 (fig). The value of $\frac{V_a}{V_d}$ will be



(A) $\frac{V_b}{V_c}$

(B) $\frac{V_c}{V_b}$

(C) $\frac{V_d}{V_a}$

(D) $V_b V_c$

Q.3

A refrigerator with coefficient of performance $\frac{1}{3}$ releases 200 J of heat to a hot reservoir. Then the work done on the working substance is

(A) $\frac{100}{3}$ J

(B) 100 J

(C) $\frac{200}{3}$ J

(D) 150 J

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Q.4

A Carnot engine takes 3×10^6 cal. of heat from a reservoir at 627°C, and gives it to a sink at 27°C. The work done by the engine is

(A) 4.2×10^6 J

(B) 8.4×10^6 J

(C) 16.8×10^6 J

(D) zero

Q.5

A diatomic ideal gas is used in a car engine as the working substance. If during the adiabatic expansion part of the cycle, volume of the gas increases from V to 32 V, the efficiency of the engine is

(A) 0.5

(B) 0.75

(C) 0.99

(D) 0.25

Q.6

A Carnot engine operating between temperatures T_1 and T_2 has efficiency $\frac{1}{6}$. When T_2 is lowered by 62 K its efficiency increases to $\frac{1}{3}$. Then T_1 and T_2 are, respectively

(A) 372 K and 330 K

(B) 330 K and 268 K

(C) 310 K and 248 K

(D) 372 K and 310 K

Q.7

If the co-efficient of performance of a refrigerator is 5 and operates at the room temperature 27°C, the temperature inside the refrigerator is

(A) 240 K

(B) 250 K

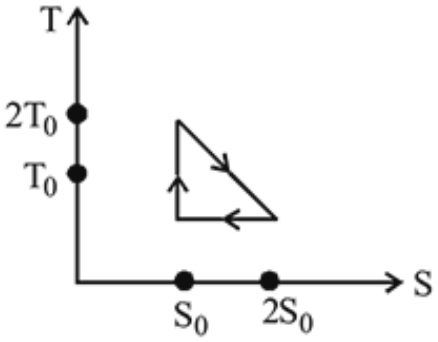
(C) 230 K

(D) 260 K

Q.8

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The temperature-entropy diagram of a reversible engine cycle is given in the figure. Its efficiency is



(A) $\frac{1}{4}$

(B) $\frac{1}{2}$

(C) $\frac{2}{3}$

(D) $\frac{1}{3}$

Q.9

A Carnot engine whose efficiency is 50% has an exhaust temperature of $500\ \mathrm{K}$. If the efficiency is to be 60% with the same intake temperature, the exhaust temperature must be (in K)

(A) 800

(B) 200

(C) 400

(D) 600

Q.10

A Carnot engine is working between $127^{\circ}\mathrm{C}$ and $27^{\circ}\mathrm{C}$. The increase in efficiency will be maximum when the temperature of

(A) the source is increased by $50^{\circ}\mathrm{C}$

(B) the sink is decreased by $50^{\circ}\mathrm{C}$

(C) source is increased by $25^{\circ}\mathrm{C}$ and that of sink is decreased by $25^{\circ\circ}\mathrm{C}$

(D) both source and sink are decreased by $25^{\circ}\mathrm{C}$ each.

Q.11

If the energy input to a Carnot engine is thrice the work it performs then, the fraction of energy rejected to the sink is

(A) $\frac{1}{3}$

(B) $\frac{1}{4}$

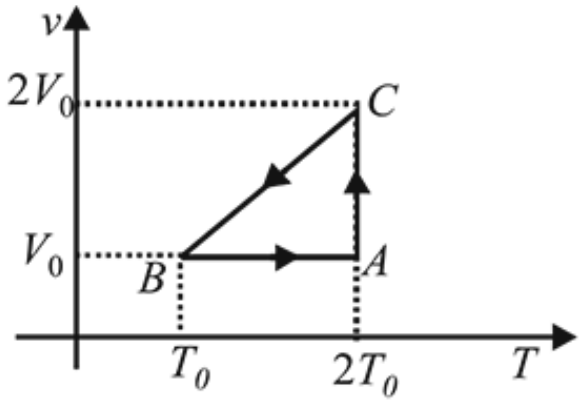
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(C) $\frac{2}{5}$

(D) $\frac{2}{3}$

Q.12

The efficiency of an ideal gas with adiabatic exponent ' γ ' for the shown cyclic process would be



(A) $\frac{(2 \ln 2-1)}{\gamma /(\gamma-1)}$

(B) $\frac{(1-2 \ln 2)}{\gamma /(\gamma-1)}$

(C) $\frac{(2 \ln 2+1)}{\gamma /(\gamma-1)}$

(D) $\frac{(2 \ln 2-1)}{\gamma /(\gamma+1)}$

Q.13

In a Carnot engine, the temperature of reservoir is $927^{\circ} \mathrm{C}$ and that of sink is $27^{\circ} \mathrm{C}$. If the work done by the engine when it transfers heat from reservoir to sink is $12.6 \times 10^6 \mathrm{~J}$, the quantity of heat absorbed by the engine from the reservoir is

(A) $16.8 \times 10^6 \mathrm{~J}$

(B) $4 \times 10^6 \mathrm{~J}$

(C) $7.6 \times 10^6 \mathrm{~J}$

(D) $4.2 \times 10^6 \mathrm{~J}$

Q.14

In Carnot engine efficiency is 40% at hot reservoir temperature T. For efficiency 50% what will be temperature of hot reservoir?

(A) $\frac{T}{5}$

(B) $\frac{2 \mathrm{~T}}{5}$

(C) $6 \mathrm{~T}$

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(D) $\frac{6\,T}{5}$

Q.15

If a refrigerator's door is opened, then we get

(A) Room heated

(B) Room cooled

(C) More amount of heat is passed out

(D) No effect on room

Answers & Solutions

Q.1 Answer:

the room will become slightly warmer

Solution:

The working of an air coditioner is similar to the working of a refrigerator. An air conditioner removes heat from the room, does some work and rejects the heat to the surroundings. As air conditioner is put in the middle of the room then due to continuous, external work the room will become slightly warmer.

Q.2 Answer:

$\frac{V_b}{V_c}$

Solution:

We know that $\frac{V_a}{V_b} = \frac{V_d}{V_c} \Rightarrow \frac{V_a}{V_d} = \frac{V_b}{V_c}$

Q.3 Answer:

150 J

Solution:

The coefficient of performance of a refrigerator is given by

$$\alpha = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2}$$

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Substituting the given values, we get

$$\frac{1}{3}=\frac{Q_2}{200-Q_2}$$

$$\Rightarrow 200-Q_2=3Q_2 \Rightarrow 4Q_2=200$$

$$\text{or } Q_2=\frac{200}{4}\ \text{J}=50\ \text{J}$$

$$\therefore \quad W=Q_1-Q_2=200\ \text{J}-50\ \text{J}=150\ \text{J}$$

Q.4 Answer:

8.4×10^6 J

Solution:

$$\eta=\frac{(627+273)-(273+27)}{627+273}$$

$$=\frac{900-300}{900}=\frac{600}{900}=\frac{2}{3}$$

$$\text{work}=(\eta)\times\text{ Heat }=\frac{2}{3}\times3\times10^6\times4.2\ \text{J}$$

$$=8.4\times10^6\ \text{J}$$

Q.5 Answer:

0.75

Solution:

The efficiency of cycle is $\eta=1-\frac{T_2}{T_1}$ **For adiabatic process** $TV^{\gamma-1}=$ **constant For diatomic gas**

$\gamma=\frac{7}{5}\ T_1V_1^{\gamma-1}=T_2V_2^{\gamma-1}\quad T_1=T_2\left(\frac{V_2}{V_1}\right)^{\gamma-1}$

$T_1=T_2(32)^{\frac{7}{5}-1}=T_2\left(2^5\right)^{2/5}=T_2\times4$

$T_1=4T_2$

$\therefore \quad \eta=\left(1-\frac{1}{4}\right)=\frac{3}{4}=0.75$

Q.6 Answer:

372 K and 310 K

Solution:

$\eta_1=1-\frac{T_2}{T_1}\Rightarrow\frac{1}{6}=1-\frac{T_2}{T_1}\Rightarrow\frac{T_2}{T_1}=\frac{5}{6}\ldots$

$\eta_2=1-\frac{T_2-62}{T_1}\Rightarrow\frac{1}{3}=1-\frac{T_2-62}{T_1}\ldots$

On solving Eqs. (i) and (ii) $T_1=372$ K **and** $T_2=310$ K

Q.7 Answer:

250 K

Solution:

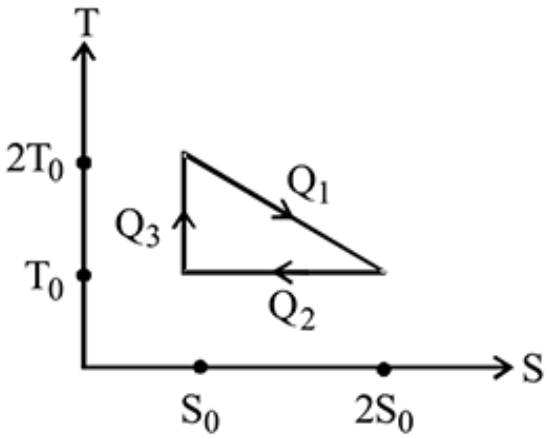
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Here, Coefficient of performance $(\beta) = 5\ \mathrm{T_1} = 27^\circ\mathrm{C} = (27 + 273)\mathrm{K} = 300\ \mathrm{K}$
As, $\beta = \frac{\mathrm{T_2}}{\mathrm{T_1} - \mathrm{T_2}} \Rightarrow 5 = \frac{\mathrm{T_2}}{300 - \mathrm{T_2}}$
or $1500 - 5\ \mathrm{T_2} = \mathrm{T_2}$ **or** $6\ \mathrm{T_2} = 1500$
$\therefore\ \mathrm{T_2} = \frac{1500}{6} = 250\ \mathrm{K}$

Q.8 Answer:

$\frac{1}{2}$

Solution:



$\mathrm{Q_1} = \mathrm{T_0\ S_0} + \frac{1}{2}\ \mathrm{T_0\ S_0} = \frac{3}{2}\ \mathrm{T_0\ S_0}$
$\mathrm{Q_2} = \mathrm{T_0}\ (2\ \mathrm{S_0} - \mathrm{S_0}) = \mathrm{T_0\ S_0}$ **and** $\mathrm{Q_3} = 0$
$\eta = \frac{\mathrm{W}}{\mathrm{Q_1}} = \frac{\mathrm{Q_1} - \mathrm{Q_2}}{\mathrm{Q_1}}$
$= 1 - \frac{\mathrm{Q_2}}{\mathrm{Q_1}} = 1 - \frac{\mathrm{T_0\ S_0}}{\frac{3}{2}\ \mathrm{T_0\ S_0}} = \frac{1}{3}$

Q.9 Answer:

400

Solution:

$\eta = 1 - \frac{T_2}{T_1}$ **or** $\frac{50}{100} = 1 - \frac{500}{T_1} \Rightarrow T_1 = 1000\ \mathrm{K}$
Also, $\frac{60}{100} = 1 - \frac{T_2}{1000} \Rightarrow T_2 = 400\ \mathrm{K}$

Q.10 Answer:

the sink is decreased by $50^\circ\mathrm{C}$

Solution:

$\frac{\mathrm{T_1} - \mathrm{T_2}}{\mathrm{T_1}}$ **is maximum in case (b).**

Q.11 Answer:

$\frac{2}{3}$

Solution:

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Efficiency $\eta = \frac{\mathrm{W_{output}}}{\mathrm{Heat\ _{input}}} = \frac{\mathrm{w}}{3\mathrm{w}} = \frac{1}{3}$

$\eta = 1 - \frac{\mathrm{Q_2}}{\mathrm{Q_1}} = \frac{1}{3} \quad \therefore \frac{\mathrm{Q_2}}{\mathrm{Q_1}} = \frac{2}{3}$

Q.12 Answer:

$\frac{(2\ln 2 - 1)}{\gamma/(\gamma-1)}$

Solution:

$W_{AB} = 0, W_{BC} = P\Delta V = nR\Delta T = -nRT_0$

$W_{CA} = nRT\ell n\frac{V_f}{V_i} = nR\left(2T_0\right)\ell n2$

$Q_{BC} = nC_p\Delta T = \left(\frac{nR\gamma}{\gamma-1}\right)T_0$

Efficiency, $\eta = \frac{W}{Q} = \left[\frac{2\ell n2-1}{\gamma/(\gamma-1)}\right]$

Q.13 Answer:

16.8×10^6 J

Solution:

As we know $\eta = \frac{\mathrm{W}}{\mathrm{Q_1}} = 1 - \frac{\mathrm{T_2}}{\mathrm{T_1}}$

$\Rightarrow \quad \eta = 1 - \frac{300\ \mathrm{K}}{1200\ \mathrm{K}} = \frac{3}{4}$

$\frac{3}{4} = \frac{\mathrm{W}}{\mathrm{Q_1}} \Rightarrow \mathrm{Q_1} = \mathrm{W} \times \frac{4}{3} \Rightarrow \mathrm{Q_1} = 12.6 \times 10^6 \times \frac{4}{3}$

$Q_1 = 16.8 \times 10^6$ J

Q.14 Answer:

$\frac{6\ \mathrm{T}}{5}$

Solution:

$\eta = 1 - \frac{T_2}{T_1}$

$\mathrm{T_1} = \mathrm{T}$ **(Temperature of hot reservoir) For** $\eta = 40\%$

$\frac{40}{100} = 1 - \frac{T_2}{T} \Rightarrow \frac{T_2}{T} = \frac{3}{5} \Rightarrow T_2 = \frac{3}{5}T$

For $\eta = 50\%$,

$\frac{50}{100} = 1 - \frac{\frac{3}{5}T}{T_1} \Rightarrow T_1 = \frac{6T}{5}$

Q.15 Answer:

Room heated

Solution:

The compressor has to run for longer time releasing more heat to the surroundings.

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